EXP 7	A PYTHON PROGRAM TO IMPLEMENT THE UNCONSTRAINED OPTIMIZATION-NONLINEAR LEAST SQUARES

AIM:

The objective of this lab is to understand and implement optimization techniques

for solving nonlinear least squares problems without constraints.

PROCEDURE:

a. Problem Formulation:

● Define the nonlinear least squares objective function f(x)f(x)f(x).

● Specify the data points and the model that relates to the data.

b. Choose an Optimization Library:

● Select an appropriate optimization library (e.g., scipy.optimize in Python).

c. Implementing the Optimization:

● Set up the objective function to be minimized.

● Define any constraints (if the problem extends to constrained NLS).

d. Running Optimization:

● Execute the chosen optimization algorithm.

● Monitor convergence and iterate as necessary.

e. Post-optimization Analysis:

● Evaluate and interpret the results obtained from the optimization.

● Compare with initial assumptions and data.

PROBLEM 1: To estimate the parameters of an exponential decay model using nonlinear least squares.

PARAMETER	VALUES
t	Linearly spaced values from 0 to 2 (60 points)
y_true	[3.0 , 0.8]
x 0(initial guess)	[2.0 , 0.3]
Noise	0.5*np.random.randn(60)
Model	X[0] * np.exo(-x[1] * t)

CODE:

```
import numpy as np
from scipy.optimize import least_squares
def model(X, t):
    return X[0] * np.exp(-X[1] * t)
def residual(X, t, y):
    return model(X, t) - y
# Generate data
np.random.seed(42)
t = np.linspace(0, 2, 60)
true_params = [3.0, 0.8]
y = model(true_params, t) + 0.05 * np.random.randn(60)
# Initial guess
x0 = [2.0, 0.3]
# Optimization
result = least_squares(residual, x0, args=(t, y))
```

print("Optimized parameters:", result.x)

OUTPUT:

Optimized parameters: [2.98 0.79]

PROBLEM 2: TO ESTIMATE THE PARAMETERS FOR A SUM OF TWO SINE WAVES USING NONLINEAR LEAST SQUARES.

PARAMETER	VALUE
t	Linearly spaced values from 0 to 10(200 points)
Y_true	[2.0,0.5,1.0,2.0,2.5]
X0(initial guess)	[1.0,1.0,1.0,1.0,1.0,1.0]
Noise	0.1*np.random.randn(200)*
Model	X[0]*sin(x[1]*t + x[2]) + x[3]*sin(x[4]*t+x[5])

CODE:

import numpy as np from scipy.optimize

import least_squares def model(X, t):

return X[0] * np.sin(X[1] * t + X[2]) + X[3] * np.sin(X[4] * t + X[5])

def residual(X, t, y):

return model(X, t) - y

np.random.seed(789) t = np.linspace(0, 10, 200)

y_true = [2.0, 0.5, 1.0, 1.0, 2.0, 2.5] y =

model(y_true, t) + 0.1 * np.random.randn(200) x0

= [1.0, 1.0, 1.0, 1.0, 1.0, 1.0]

result = least_squares(residual, x0, args=(t, y))

print("optimized parameters:", result.x)

OUTPUT:

optimized parameters: [0.97587131 0.54470271 0.77416988 0.97586095 0.54475034

0.77408607

RESULT:

Thus, solving any Unconstrained Optimization using the Non-Linear Least Squares Algorithm in python has been implemented and executed successfully